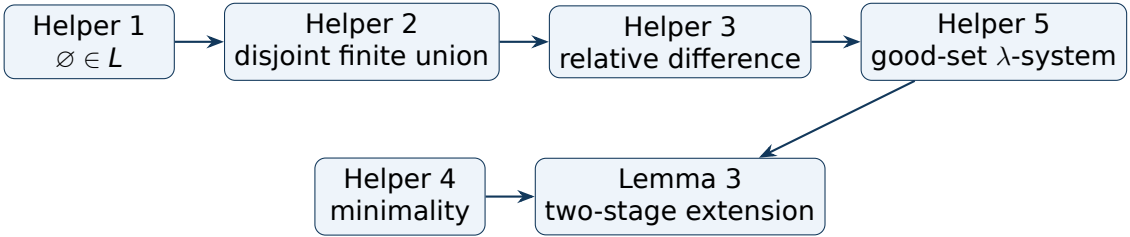
Project A: The π–λ Theorem

Lemmas 1–4, Helper Lemmas 1–5, and the final theorem

Math 7800 · Project A · Oral examination review

Fall 2026

Proof roadmap: how the helpers fit together



Background used in Lemma 3

Lemma 1: $\lambda(\mathcal{P})$ is a λ-system. Lemma 2: $\mathcal{P} \subseteq \lambda(\mathcal{P})$.

Lemma 1: Generated λ-system is a λ-system

Statement

For any class $\mathcal{A} \subseteq \mathcal{P}(\Omega)$, the generated class $\lambda(\mathcal{A})$ is a λ-system.

Idea

It is the intersection of all λ-systems containing $\mathcal{A}$. Each of the three defining properties holds in every eligible L, hence in their intersection.

Actual Lean proof

```
lemma lambdaGen_isLambdaSys (A : Set (Set Ω)) :
    IsLambdaSys (lambdaGen A) := by
  refine ⟨?_, ?_, ?_⟩
  · exact mem_sInter.2 fun L hL => hL.1.univ_mem
  · intro B hB
    exact mem_sInter.2 fun L hL =>
      hL.1.compl_mem B ((mem_sInter.1 hB) L hL)
  · intro f hd hf
    exact mem_sInter.2 fun L hL =>
      hL.1.iUnion_mem f hd
        (fun n => mem_sInter.1 (hf n) L hL)
```

Lemma 1: Final tactic state

Final tactic state (immediately before the final tactic)

$f : \mathbb{N} \to \mathrm{Set}(\Omega)$, $hd : \mathrm{Pairwise}(\mathrm{Disjoint} \circ f)$,
$hf : \forall n,\ f(n) \in \lambda(\mathcal{A})$;
Goal: $\bigcup_n f(n) \in \lambda(\mathcal{A})$.

Final tactic

```
exact mem_sInter.2 fun L hL =>
      hL.1.iUnion_mem f hd (fun n => ... )
```

$\Longrightarrow$ **No goals**

Why it closes: For each eligible L, its countable disjoint-union property applies to the sequence whose entries all belong to L.

Lemma 2: The generators belong to their generated class

Statement

For every class $\mathcal{A}$, we have $\mathcal{A} \subseteq \lambda(\mathcal{A})$.

1. Fix $B \in \mathcal{A}$.
2. Every eligible λ-system L contains $\mathcal{A}$, so $B \in L$.
3. Hence B belongs to their intersection, $\lambda(\mathcal{A})$.

Actual Lean proof

```
lemma subset_lambdaGen (A : Set (Set Ω)) :
    A ⊆ lambdaGen A := by
  intro B hB
  exact mem_sInter.2 fun L hL => hL.2 hB
```

Lemma 2: Final tactic state

Final tactic state (immediately before the final tactic)

$B : \mathrm{Set}(\Omega)$, $hB : B \in \mathcal{A}$;
Goal: $B \in \lambda(\mathcal{A})$.

Final tactic

```
exact mem_sInter.2 fun L hL => hL.2 hB
```
$\Longrightarrow$ **No goals**

Why it closes: $hL.2$ is the inclusion $\mathcal{A} \subseteq L$ for an arbitrary eligible L.

Helper 1: Empty set belongs to a λ-system

Mathematical statement

If L is a λ-system, then $\varnothing \in L$.

1. By definition, $\Omega \in L$.
2. Complement closure gives $\Omega^c \in L$.
3. Since $\Omega^c = \varnothing$, conclude $\varnothing \in L$.

Key Lean code

```
have h_univ : (univ : Set Ω) ∈ L := hL.univ_mem
have h_compl : (univ : Set Ω)ᶜ ∈ L :=
  hL.compl_mem univ h_univ
simpa using h_compl
```

Helper 1: Final tactic state

Final tactic state (immediately before the final tactic)

$L : \mathrm{Set}(\mathrm{Set}\,\Omega)$, $hL : \mathrm{IsLambdaSys}(L)$,
$h_compl : (\Omega)^c \in L$
Goal: $\varnothing \in L$.

Final tactic

`simpa using h_compl` $\Longrightarrow$ **No goals**

Why it closes: Lean simplifies the complement of the universal set to the empty set.

Helper 2: Finite disjoint unions

Statement

If $A, B \in L$ and $A \cap B = \varnothing$, then $A \cup B \in L$.

1. Build a sequence $f(0) = A$, $f(1) = B$, $f(n) = \varnothing$ for $n \geq 2$.
2. Show every $f(n) \in L$ (Helper 1 handles the remaining terms).
3. Prove f is pairwise disjoint (use $A \perp B$ and symmetry).
4. Prove $\bigcup_n f(n) = A \cup B$ by two inclusions.
5. Apply the λ-system's countable disjoint-union property.

Core Lean evidence

```
have h_big_union : (⋃ n, f n) ∈ L :=
  hL.iUnion_mem f hf_disjoint hf_mem
rw [h_union] at h_big_union
exact h_big_union
```

Helper 2: Final tactic state

Final tactic state (immediately before the final tactic)

h_union : $(\bigcup_n f(n)) = A \cup B$,
h_big_union : $A \cup B \in L$ *(after rewriting)*
Goal: $A \cup B \in L$.

Final tactic

`exact h_big_union` $\Longrightarrow$ **No goals**

Why it closes: The rewritten hypothesis exactly matches the goal.

Helper 3: Relative differences

Statement

If $A, B \in L$ and $A \subseteq B$, then $B \setminus A \in L$.

1. $A \subseteq B$ implies $A \cap B^c = \varnothing$.
2. $B^c \in L$ by complement closure.
3. Helper 2 gives $A \cup B^c \in L$.
4. Complement closure gives $(A \cup B^c)^c \in L$.
5. De Morgan's law: $(A \cup B^c)^c = B \setminus A$.

Core Lean evidence

```
have hunion : A ∪ Bᶜ ∈ L :=
  lambda_union_mem_of_disjoint L hL A Bᶜ
    hA (hL.compl_mem B hB) hdisj
have hcompl : (A ∪ Bᶜ)ᶜ ∈ L :=
  hL.compl_mem (A ∪ Bᶜ) hunion
rw [heq] at hcompl
exact hcompl
```

Helper 3: Final tactic state

Final tactic state (immediately before the final tactic)

$h_eq : (A \cup B^c)^c = B \setminus A$,
$h_compl : B \setminus A \in L$ *(after rewriting)*
Goal: $B \setminus A \in L$.

Final tactic

`exact hcompl` $\Longrightarrow$ **No goals**

Why it closes: The set identity turns complement closure into relative-difference closure.

Helper 4: Minimality of the generated λ-system

Statement

If L is a λ-system and $\mathcal{P} \subseteq L$, then $\lambda(\mathcal{P}) \subseteq L$.

Main idea

By definition, $\lambda(\mathcal{P})$ is the intersection of *all* λ-systems containing $\mathcal{P}$. Since L is one such eligible system, every $E \in \lambda(\mathcal{P})$ is also in L.

Lean proof

```
intro E hE
exact (Set.mem_sInter.mp hE) L (hL, hPL)
```

`Set.mem_sInter.mp hE`: membership in an intersection means membership in every eligible family member.

Helper 4: Final tactic state

Final tactic state (immediately before the final tactic)

$hE : E \in \lambda(\mathcal{P})$,
$hL : \mathrm{IsLambdaSys}(L)$, $hPL : \mathcal{P} \subseteq L$
Goal: $E \in L$.

Final tactic

```
exact (Set.mem_sInter.mp hE) L ⟨hL, hPL⟩
```
$\Longrightarrow$ **No goals**

Why it closes: The pair $\langle hL, hPL \rangle$ certifies that L is eligible.

Helper 5: The good-sets construction

Statement

If L is a λ-system and $D \in L$, define

$$\mathcal{G}_D = \{B \subseteq \Omega : D \cap B \in L\}.$$

Then $\mathcal{G}_D$ is a λ-system.

1. **Universal set:** $D \cap \Omega = D \in L$.
2. **Complements:** from $D \cap B \in L$, use Helper 3 to obtain
$$D \setminus (D \cap B) = D \cap B^c \in L.$$
3. **Disjoint unions:** for pairwise disjoint $f(n) \in \mathcal{G}_D$, the sets $D \cap f(n)$ are pairwise disjoint and belong to L; then
$$\bigcup_n (D \cap f(n)) = D \cap \bigcup_n f(n) \in L.$$

Helper 5: Essential Lean tactics

```
refine {
  univ_mem := by
    change D ∩ univ ∈ L
    simpa only [inter_univ] using hD
  compl_mem := by
    intro B hB
    change D ∩ B ∈ L at hB
    change D ∩ Bᶜ ∈ L
    have hd : D \ (D ∩ B) ∈ L :=
      lambda_diff_mem L hL
        (D ∩ B) D hB hD inter_subset_left
    -- heq : D \ (D ∩ B) = D ∩ Bᶜ
    rw [heq] at hd
    exact hd
  iUnion_mem := by
    intro f hdisj hf
    change D ∩ (⋃ n, f n) ∈ L
    -- hd: Pairwise (Disjoint on (fun n => D ∩ f n))
    have hu : (⋃ n, D ∩ f n) ∈ L :=
      hL.iUnion_mem (fun n => D ∩ f n) hd hf
    -- heq : (⋃ n, D ∩ f n) = D ∩ (⋃ n, f n)
    rw [heq] at hu
    exact hu
}
```

Omitted internal proofs of `hd` and `heq` are shown conceptually on the preceding slide; this is an excerpt, not standalone Lean code.

Helper 5: Final tactic state

Final tactic state (immediately before the final tactic)

$hu : D \cap \left(\bigcup_n f(n)\right) \in L$ *(after rewriting)*
Goal: $D \cap \left(\bigcup_n f(n)\right) \in L$.

Final tactic

`exact hu` $\Longrightarrow$ **No goals**

Why it closes: This completes the third field, `iUnion_mem`. The closing brace ends the IsLambdaSys structure proof.

Lemma 3: Statement and why two stages are needed

Statement

If $\mathcal{P}$ is a π-system, then

$$A, B \in \lambda(\mathcal{P}) \quad \Longrightarrow \quad A \cap B \in \lambda(\mathcal{P}).$$

Obstacle

The π-system hypothesis only gives intersection closure for *both* inputs in $\mathcal{P}$. We need both inputs in the larger class $\lambda(\mathcal{P})$.

1. **Stage 1:** $D \in \mathcal{P}$, $B \in \lambda(\mathcal{P}) \implies D \cap B \in \lambda(\mathcal{P})$.
2. **Stage 2:** $A, B \in \lambda(\mathcal{P}) \implies A \cap B \in \lambda(\mathcal{P})$.

Lemma 3, Stage 1: Extend the second argument

Fix $D \in \mathcal{P}$ and define the good sets

$$\mathcal{G}_D = \{B : D \cap B \in \lambda(\mathcal{P})\}.$$

1. Lemma 1 shows $\lambda(\mathcal{P})$ is a λ-system; Lemma 2 gives $D \in \lambda(\mathcal{P})$.
2. Helper 5 shows $\mathcal{G}_D$ is a λ-system.
3. If $B \in \mathcal{P}$, the π-system property gives $D \cap B \in \mathcal{P} \subseteq \lambda(\mathcal{P})$. Thus $\mathcal{P} \subseteq \mathcal{G}_D$.
4. Helper 4 gives $\lambda(\mathcal{P}) \subseteq \mathcal{G}_D$. Therefore every $B \in \lambda(\mathcal{P})$ satisfies $D \cap B \in \lambda(\mathcal{P})$.

Essential Lean code

```
have hgood :
    IsLambdaSys {B | D ∩ B ∈ lambdaGen P} :=
  lambdaSys_good (lambdaGen P) hK D (hPK hD)
-- hcontains: P ⊆ {B | D ∩ B ∈ lambdaGen P}
exact lambdaGen_min P
  {B | D ∩ B ∈ lambdaGen P} hgood hcontains
```

Lemma 3, Stage 2: Extend the first argument

Fix $A \in \lambda(\mathcal{P})$ and define a second good-set family

$$\mathcal{G}_A = \{D : A \cap D \in \lambda(\mathcal{P})\}.$$

1. Helper 5 shows $\mathcal{G}_A$ is a λ-system.
2. For every $D \in \mathcal{P}$, Stage 1 gives $D \cap A \in \lambda(\mathcal{P})$.
3. Use $D \cap A = A \cap D$ to show $\mathcal{P} \subseteq \mathcal{G}_A$.
4. Helper 4 gives $\lambda(\mathcal{P}) \subseteq \mathcal{G}_A$. As $B \in \lambda(\mathcal{P})$, we have $B \in \mathcal{G}_A$, i.e. $A \cap B \in \lambda(\mathcal{P})$.

Essential Lean code

```
have hgood :
    IsLambdaSys {D | A ∩ D ∈ lambdaGen P} :=
  lambdaSys_good (lambdaGen P) hK A hA
-- Within hcontains:
simpa only [inter_comm] using hfirst D hD A hA
```

Lemma 3: Main proof skeleton

```
have hK : IsLambdaSys (lambdaGen P) :=
  lambdaGen_isLambdaSys P
have hPK : P ⊆ lambdaGen P :=
  subset_lambdaGen P

have hfirst :
    ∀ D ∈ P, ∀ B ∈ lambdaGen P,
      D ∩ B ∈ lambdaGen P := by
  intro D hD
  -- Build G_D; prove hgood and hcontains.
  exact lambdaGen_min P
    {B | D ∩ B ∈ lambdaGen P} hgood hcontains

intro A hA B hB
-- Build G_A; prove hgood and hcontains.
have hsub :
    lambdaGen P ⊆ {D | A ∩ D ∈ lambdaGen P} :=
  lambdaGen_min P
    {D | A ∩ D ∈ lambdaGen P} hgood hcontains
exact hsub hB
```

Condensed to foreground proof architecture; omitted `have` subproofs are detailed on the preceding slides.

Lemma 3: Final tactic state

Final tactic state (immediately before the final tactic)

$hA : A \in \lambda(\mathcal{P})$, $hB : B \in \lambda(\mathcal{P})$,
$hsub : \lambda(\mathcal{P}) \subseteq \{D : A \cap D \in \lambda(\mathcal{P})\}$
Goal: $A \cap B \in \lambda(\mathcal{P})$.

Final tactic

`exact hsub hB` $\Longrightarrow$ **No goals**

Why it closes: Applying inclusion *hsub* to *hB* places B in the good-set family, whose defining property is precisely the goal.

Where Lemmas 1–3 fit

Prerequisites for Lemma 3

Lemma 1: $\lambda(\mathcal{P})$ is itself a λ-system (needed to invoke Helper 5).

Lemma 2: $\mathcal{P} \subseteq \lambda(\mathcal{P})$ (needed to convert $D \in \mathcal{P}$ into $D \in \lambda(\mathcal{P})$).

Lemma 3: When $\mathcal{P}$ is a π-system, $\lambda(\mathcal{P})$ is also closed under intersections.

What comes next?

Lemma 4 combines λ-system properties with finite intersection closure to obtain a σ-field. The final theorem uses the minimality of the generated σ-field.

Lemma 4: A λ-system closed under $\cap$ is a σ-field

Statement

If L is a λ-system and $A, B \in L \Rightarrow A \cap B \in L$, then L is a σ-field.

1. Universal-set and complement closure come directly from *hL*.
2. To prove arbitrary countable union closure, rewrite $\bigcup_n f(n)$ as the union of its *disjointed* sequence.
3. Each piece is built by repeated differences $t \setminus f(i) = t \cap f(i)^c$, still in L.
4. Apply *hL.iUnion_mem* to the now pairwise-disjoint pieces.

Lemma 4: Key Lean tactics

Disjointification and relative differences

```
refine ⟨?_, ?_, ?_⟩
· exact hL.univ_mem
· exact hL.compl_mem
· intro f hf
  rw [← iUnion_disjointed]
  apply hL.iUnion_mem (disjointed f)
    (disjoint_disjointed f)
  intro n
  refine disjointedRec (p := (· ∈ L)) ?_ (hf n)
  intro t i ht
  rw [sdiff_eq]
  exact hm t ht (f i)ᶜ
    (hL.compl_mem (f i) (hf i))
```

Lemma 4: Final tactic state

Final tactic state (immediately before the final tactic)

$h\pi : \forall A, B \in L,\ A \cap B \in L,$
$ht : t \in L, \quad hf : \forall i,\ f(i) \in L;$
Goal: $t \cap f(i)^c \in L$ *(after* `rw [sdiff_eq]`*)*.

Final tactic

```
exact hπ t ht (f i)ᶜ
      (hL.compl_mem (f i) (hf i))
```
$\Longrightarrow$ **No goals**

Why it closes: The complement $f(i)^c$ belongs to L; intersection closure completes the recursive disjointification step.

Final theorem: The π–λ theorem

Statement

If $\mathcal{P}$ is a π-system and L is a λ-system with $\mathcal{P} \subseteq L$, then

$$\boxed{\sigma(\mathcal{P}) \subseteq L.}$$

1. Minimality gives $\lambda(\mathcal{P}) \subseteq L$.
2. Lemma 3 makes $\lambda(\mathcal{P})$ a π-system.
3. Lemma 1 + Lemma 4 make $\lambda(\mathcal{P})$ a σ-field.
4. Lemma 2 says $\mathcal{P} \subseteq \lambda(\mathcal{P})$; therefore $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$.
5. Chain the inclusions to obtain $\sigma(\mathcal{P}) \subseteq L$.

Final theorem: Actual Lean proof structure

```
theorem pi_lambda (P L : Set (Set Ω))
    (hP : IsPiSys P) (hL : IsLambdaSys L)
    (hPL : P ⊆ L) : sigmaGen P ⊆ L := by
  have hGenLambda : lambdaGen P ⊆ L := by
    intro A hA
    unfold lambdaGen at hA
    exact Set.mem_sInter.mp hA L ⟨hL, hPL⟩

  have hGenPi : IsPiSys (lambdaGen P) := by
    exact lambdaGen_inter P hP

  have hGenSigma : IsSigmaField (lambdaGen P) := by
    exact isSigmaField_of_lambda_pi
      (lambdaGen P) (lambdaGen_isLambdaSys P) hGenPi

  have hSigmaSubset : sigmaGen P ⊆ lambdaGen P := by
    intro A hA
    unfold sigmaGen at hA
    exact Set.mem_sInter.mp hA (lambdaGen P)
      ⟨hGenSigma, subset_lambdaGen P⟩

  exact Set.Subset.trans hSigmaSubset hGenLambda
```

Final theorem: Final tactic state

Final tactic state (immediately before the final tactic)

hSigmaSubset : $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$,
hGenLambda : $\lambda(\mathcal{P}) \subseteq L$;
Goal: $\sigma(\mathcal{P}) \subseteq L$.

Final tactic

```
exact Set.Subset.trans hSigmaSubset hGenLambda
```
$\Longrightarrow$ **No goals**

Why it closes: Subset inclusion is transitive; the generated λ-system is the intermediate collection.

Oral exam: questions across Project A

1. **Why are good sets needed?** To construct a λ-system on which minimality can extend the intersection property.
2. **Why use minimality twice?** Once to extend B from $\mathcal{P}$ to $\lambda(\mathcal{P})$, and again to extend A from $\mathcal{P}$ to $\lambda(\mathcal{P})$.
3. **Why is Helper 3 needed?** Good-set complement closure requires the relative difference $D \setminus (D \cap B)$.
4. **What does `Set.mem_sInter.mp` mean?** Membership in every eligible set of the intersection.
5. **What does `exact hsub hB` do?** Uses subset membership as a function to derive $B \in \mathcal{G}_A$.